

AANCHAL SAVE

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EDUCATION

Georgia Institute of Technology

B.S. Aerospace Engineering; Minor: Computer Science; GPA: 3.9/4.0

Atlanta, GA

May 2028

EXPERIENCE

Anduril AI Grand Prix Team (Georgia Tech Representative)

Atlanta, GA

Autonomous Control and Decision Systems (ACDS) Laboratory

Apr 2026 – Present

- Selected as 1 of 8 Georgia Tech representatives; **placed 3rd of 3,000+ teams** virtually, advancing to hardware qualifiers.
- Developing monocular autonomy with YOLO/CNN gate detection, GTSAM state estimation, and MPC/MPPI control.
- Building gate-pose estimation from detected corners and camera geometry to support high-speed trajectory planning.
- Evaluating perception and control across simulation and hardware to identify and reduce sim-to-real failure modes.

Software Engineering Intern – Edge AI Pipelines

New York, NY

Palantir Technologies

May 2026 – Aug 2026

- Enabled stateful/stateless user functions in real-time edge pipelines via sidecar containers and JVM classloading.
- Designed sidecar execution around security isolation and performance constraints on compute- and memory-limited hardware.
- Built across Kubernetes, Flink, Docker, and Cassandra to support distributed execution and real-time sensor processing.
- Reduced cloud dependency by enabling user-defined processing to execute alongside streaming pipelines at the edge.

GNC Software Engineering Intern

Hawthorne, CA

SpaceX

Sep 2025 – Dec 2025

- Designed Starshield constellation architectures across deployment, coverage, and cost trade spaces.
- Built C++/Python constellation simulations, performance metrics, and visualizations for architecture trades.
- Led trade discussions and presented analyses to senior leadership, supporting negotiations for a **\$4.16B government contract**.

Machine Learning and Robotics Intern

Huntsville, AL

Amentum (NASA MSFC EV-43)

May 2025 – Aug 2025

- Trained PyTorch forward/inverse kinematics networks for a 7-DOF Kinova arm on 100k+ simulated poses.
- Generated training data in Gazebo/Simulink and iteratively validated kinematic prediction error.
- **Accepted for a NASA New Technology Report (NTR)** for a novel inverse kinematics method under patent consideration.
- Built a ROS2 robot-control stack integrating YOLOv11, LLaMA, Whisper, Chainlit, RAG, and speech input.

Software and Mechanical Team Member

Atlanta, GA

RoboJackets – RoboNav (University Rover Challenge)

Aug 2024 – May 2025

- Built a ROS2/C++ LIDAR pipeline converting point-cloud terrain geometry into navigation cost maps.
- Computed slope, roughness, and elevation features and fused terrain costs into autonomous navigation.

RESEARCH EXPERIENCE

Undergraduate Research Assistant

Atlanta, GA

DREAM Lab; PI: Dr. Cédric Pradalier, Dr. Stéphanie Aravecchia

Jan 2026 – May 2026

- Evaluated TD-MPC2, PPO, and PID in MuJoCo and on TurtleBot2, achieving 100% sim-to-real transfer.
- Fixed degenerate PPO spinning through reward redesign and observed emergent planning with TD-MPC2.
- Applied curriculum learning and domain randomization to maintain robustness under a 1 kg payload.

Undergraduate Research Assistant

Atlanta, GA

CEREAL Lab; PI: Dr. Juergen Rauleder

Nov 2024 – May 2025

- Built Bayesian PyTorch models predicting airfoil pressure distributions and uncertainty from flight conditions.
- Built an LLM + RAG pipeline for natural-language retrieval and extraction across aerodynamic datasets.

Undergraduate Research Assistant

Atlanta, GA

Dynamics and Controls Systems Laboratory; PI: Dr. Panagiotis Tsiotras

Aug 2024 – Dec 2024

- Built a ROS2/Python F1TENTH stack using YOLO detection and disparity-based depth estimation.
- Fused LIDAR and stereo vision for real-time obstacle detection, avoidance, and vehicle control.

PUBLICATIONS

- A. Save, “Artificial Neural Networks for 7-DOF Robot Forward and Inverse Kinematics,” *IEEE ICMRE 2026* (Oral).
- W. Lowman, A. Save, D. Ho et al., “Distributed Intelligence Architecture for Edge Robotics,” *AIAA ASCEND 2026*.
- H. Lee, A. Save, P. Seshadri et al., “Large Airfoil Models,” *Computers & Fluids*, vol. 298, 106662, 2025.
- “Exploring Atmospheric Effects During Solar Eclipses,” *AGU Fall Meeting 2023*.

SKILLS

- **Programming:** Python, C++, Java, MATLAB, JavaScript, C#
- **Robotics/ML:** ROS2, PyTorch, OpenCV, YOLO, GTSAM, Gazebo, MuJoCo, Simulink, Orekit
- **Systems:** Linux, Git, SQL, Kubernetes, Flink, Cassandra, Docker, distributed systems